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The Editors
Machine Learning
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Dear Editors,

I submit for your consideration a research article entitled **"When Conformal Fusion Cannot Be Calibrated: Attainability Limits for Trace-Based Detection of Prompt-Injection Compromise."**

The paper is a methodological contribution about combining conformal p-values, motivated and evaluated on a security problem. Language-model agents that call external tools ingest untrusted content and fold it into the context that drives their next action, so an instruction hidden in a document can be followed as though the user had issued it. Runtime monitors for this compute several trace-derived signals and merge them into one alarm. Every paper I am aware of treats the merging step as plumbing. The central result here is that it is the binding constraint, and that the constraint is a sample-size law.

Main result

A conformal p-value computed from n calibration values cannot fall below $1/(n+1)$. That floor caps the evidence any calibrator in the standard family $f_\lambda(p) = \lambda p^{\lambda-1}$ can express, at exactly e^{u-1}/u with $u = \log(n+1)$, attained at $\lambda = 1/u$. Averaging calibrated e-values is the only merging rule that remains valid when the signals are arbitrarily dependent, and an average never exceeds its largest term. The level-α dependence-robust test therefore has an *empty* rejection region—power exactly zero, on every conceivable data set—unless $e^{u-1}/u \geq 1/\alpha$. At $\alpha = 0.05$ that first holds at $n = 312$ calibration examples. A companion proposition gives the corresponding turn-on for the Bonferroni-style rules, $n \geq K/\alpha - 1$, which is 39 for two signals.

Both statements are elementary once posed, and I have not found either in the conformal-prediction or e-value literature, which studies merging functions under the assumption that the underlying p-values are continuous. Discreteness is unavoidable for split-conformal p-values, and the consequence is a hard sample-size requirement rather than a loss of efficiency.

Supporting empirical work

On 90 genuinely compromised trajectories collected from two commercial agents, eight combination rules produce pooled AUROC within 0.010 of one another while their a-priori true-positive rates span 0.000 to 0.867. Discrimination is nearly rule-independent; the ability to honour a stated false-alarm rate is almost entirely determined by the rule, which is an axis a literature reporting AUROC alone cannot see. The three dependence-robust rules behave exactly as the propositions require—the e-average, mean-p and the union bound are inert, and Bonferroni is the only one that fires, with power equal to the fraction of positives carrying one signal at the p-value floor, 0.416, to three decimals. That is an identity rather than a coincidence: the two signals are at the floor together in 0.000 of cases. Injection evidence in these traces is therefore dense in the bulk, where additive fusion ranks best, and sparse in the tail, where only a minimum-based rule can be calibrated, and those two facts select opposite rules.

The argument also explains, from one cause, a negative result the project had recorded without explanation: an anytime-valid e-process detector was outperformed by a hand-picked benign quantile. Time-uniform guarantees and dependence-robust merging both purchase safety with a multiplicative budget, and both need more calibration evidence than corpora of this size can express.

Why *Machine Learning*

The contribution is statistical methodology—conformal p-values, e-values and their combination under dependence—and its audience is readers who care about calibration and validity rather than about one application domain. The security setting supplies the motivation, a real corpus and a concrete consequence for people building such corpora, but the propositions apply to any conformal monitor that fuses several scores, which is now a common construction in machine-learning systems monitoring. The paper also reports its negative results in full: five components implemented and rejected on the evidence, a proof that the intuitive normalisation of subtracting the largest benign training value caps AUROC at one half, and a label-polarity trap in a standard evaluation harness that inverted my own conclusions for a period.

Every number, table and figure is regenerated by released scripts from released data, without any model API access, because the model-judge scores are cached and committed. A verification script re-derives each reported value and fails on any mismatch.

Declarations

The manuscript is original, has not been published previously, and is not under consideration elsewhere. As sole author I am responsible for all aspects of the work and I approve it for submission. There are no competing interests and no external funding was received. The work involves no human participants, human data or animals. All code and data are public:
`https://github.com/vinhqdang/SENTRY-Anytime-Valid-E-Process-Monitoring-of-LLM-Agent-Action-Streams`

Thank you for your consideration.

Sincerely,

Quang-Vinh Dang